

- Q.26 Explain working of thermocouple transducer.
 Q.27 Draw and explain working of VCG.
 Q.28 Differentiate between polarized and non-polarized electrodes.
 Q.29 Draw and explain the structure of human heart.
 Q.30 Write short note on pacemaker.
 Q.31 Draw block diagram of ultrasound machine.
 Q.32 Which type of electrodes are used for EMG measurements.
 Q.33 Differentiate between micro shock and macro shock.
 Q.34 Explain suction cup electrode in brief.
 Q.35 Write short note on testing of biomedical equipment.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Draw the block diagram of CT scanner and explain its working.
 Q.37 What are the various safety aspects of medical instruments.
 Q.38 Write short note on any two
 a) Skin contact impedance
 b) Normal ECG waveform
 c) Leakage current

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4th Sem / Eltx Sub : Medical Electronics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 CT stand for
 a) Computerized Tomography
 b) Computer Typing
 c) Calibration Tomography
 d) Calibration Typing
- Q.2 Full form of ECG is
 a) Electric cardiogram b) Electro cardiogram
 c) Electric cardiograph d) None
- Q.3 What are the standard number of electrodes used in ECG
 a) 4 b) 6
 c) 8 d) 10
- Q.4 The SA node stands for _____
 a) Sinoatrial node b) Sino artery node
 c) Sinusoidal node d) None

- Q.5 LVDT stands for
- Linear variable different transformer
 - Line variable differential transformer
 - Linear variable different transformer
 - Line variable different transformer
- Q.6 Which of the following is used as internal electrode
- Suction Electrode
 - Floating Electrode
 - Needle Electrode
 - Metal Plate Electrode
- Q.7 The CT scanner is an example of
- Therapeutic equipment
 - Diagnostic equipment
 - Clinical laboratory equipment
 - None of these
- Q.8 MRI stands for
- Magnetic Resonance Imaging
 - Material Resonance Imaging
 - Magnetic Resonant imaging
 - None of these
- Q.9 Beats / minute of a normal human being ranges from
- 30 to 60
 - 60 to 100
 - 100 to 120
 - 120 to 150
- Q.10 Thermocouples are used for measurement of
- Pressure
 - Resistance
 - Temperature
 - Heart beat

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is the main function of heart?
- Q.12 The operation of thermocouple is based on _____ (Doppler / See back effect)
- Q.13 What is the use of spirometer?
- Q.14 Pacemaker is used when _____ (Brain / Heart) is not working properly?
- Q.15 What is bio electrode?
- Q.16 Diastolic pressure is the minimum pressure in the arteries (True / False)
- Q.17 The frequency of muscle activity is very high (True / False)
- Q.18 What is gross shock?
- Q.19 Name any two types of temperature transducers.
- Q.20 Write two applications of EEG machine.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between EEG and EMG.
- Q.22 What is Defibrillator and what is it used for?
- Q.23 Explain metal electrolyte interface.
- Q.24 Explain various body surface electrodes.
- Q.25 What are therapeutic equipments? Name any two.